

True or false? (unless otherwise specified.)

Answer grid

	1	2	3	4	5	6	7	8	9	10
True	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
False	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

1. The degree of precision of the following quadrature rule is 1:

$$\int_{-1}^1 f(x) dx \approx 2f(0).$$

2. The closed Newton–Cotes rule with $n + 1$ points is exact for all polynomials of degree up to n .

3. The trapezoidal approximation is always wrong if f is not a polynomial. That is to say, for every non-polynomial function f , it holds that

$$\int_{-1}^1 f(x) dx \neq f(-1) + f(1).$$

4. The degree of precision of the following quadrature rule is 1:

$$\int_{-1}^1 f(x) dx \approx f(1).$$

5. Suppose that $f \in C^\infty[a, b]$ and let $I_n[f]$ denote the approximate integral of f using the composite trapezoidal rule with $n + 1$ equidistant integration points. Then it holds that

$$\lim_{n \rightarrow \infty} |I[f] - I_n[f]| = 0, \quad I[f] := \int_a^b f(x) dx.$$

6. Suppose that $f \in C^\infty[a, b]$ and let $I_n[f]$ denote the approximate integral of f using the composite trapezoidal rule with $n + 1$ equidistant integration points. Then there exists C such that

$$\forall n \in \{1, 2, \dots\}, \quad |I[f] - I_n[f]| \leq \frac{C}{n^2}.$$

7. Suppose that $f \in C^\infty[a, b]$ and let $I_n[f]$ denote the approximate integral of f using the composite trapezoidal rule with $n + 1$ equidistant integration points. Then it holds that

$$\lim_{n \rightarrow \infty} n^2 |I[f] - I_n[f]| = 0.$$

8. There exist w_1 and w_2 such that the degree of precision of the following rule is 2:

$$\int_{-1}^1 f(x) dx \approx w_1 f(0) + w_2 f(1).$$

9. In Julia, the following code implements the composite trapezoidal rule with $n + 1$ points:

```
function I_approx(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    return h*sum(f, x[1:n]) .+ h/2)
end
```

10. In Julia, the following code implements the composite Simpson rule with $n + 1$ points, where n is even:

```
function I_approx(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    result = 0.
    result += h/3 * f(x[1]) + h/3 * f(x[end])
    result += 4h/3 * sum(f, x[2:2:end-1])
    result += 2h/3 * sum(f, x[3:2:end-2])
    return result
end
```

Answer key

1. **True.** The rule is exact for 1 and x (by symmetry), but not for x^2 : $\int_{-1}^1 x^2 dx = \frac{2}{3} \neq 0 = 2 \cdot 0^2$.
2. **True.** The rule integrates exactly the interpolating polynomial of degree at most n , which coincides with f when $f \in \mathcal{P}_n$. (The degree of precision can be higher, for instance for Simpson's rule.)
3. **False.** For example $f(x) = \sin(\pi x)$ is not a polynomial, but $\int_{-1}^1 f(x) dx = 0 = f(-1) + f(1)$.
4. **False.** The rule is not even exact for constants: for $f = 1$ the exact integral is 2 but the rule gives 1, so the degree of precision is not 1 (it is not even 0).
5. **True.** For $f \in C^2[a, b]$, the error is bounded by $\frac{b-a}{12} C_2 h^2$ with $h = \frac{b-a}{n}$, which tends to zero.
6. **True.** This follows from the error bound $\frac{(b-a)^3}{12} \frac{C_2}{n^2}$.
7. **False.** The error is of order h^2 and this is sharp in general. For $f(x) = x^2$ on $[0, 1]$, the error is exactly $\frac{h^2}{6}$, so $n^2 |I[f] - I_n[f]| = \frac{1}{6}$ for all n .
8. **False.** Exactness for 1, x and x^2 would require $w_1 + w_2 = 2$, $w_2 = 0$ and $w_2 = \frac{2}{3}$, which is contradictory. (Two free weights cannot achieve degree of precision 2 here.)
9. **False.** This code implements the composite *midpoint* rule: the function is evaluated at the midpoints `x[1:n] .+ h/2` of the subintervals, not at the nodes.
10. **True.** The weights are $\frac{h}{3} \times (1, 4, 2, 4, \dots, 2, 4, 1)$: the odd-indexed interior nodes (`x[2:2:end-1]`) get weight $\frac{4h}{3}$ and the even-indexed interior nodes (`x[3:2:end-2]`) get weight $\frac{2h}{3}$.